# Certificate

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Roll No:  Exam No:

Institution _____

*This is certified to be the bonafide work of the student in the* _____ *Laboratory during the academic year 20 / 20 .*

*No. of practicals certified* _____ *out of* _____ *in the subject of* _____

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*(N.B: The candidate is expected to retain his/her journal till he/she passes in the subject.)*

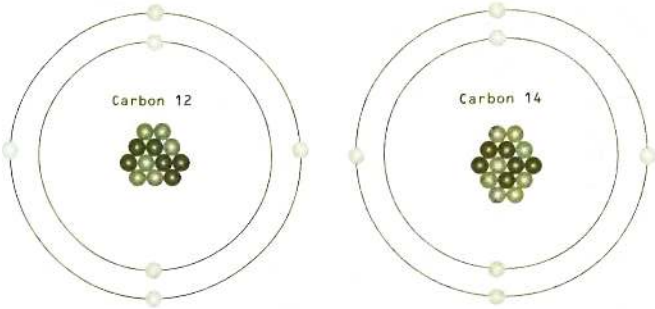
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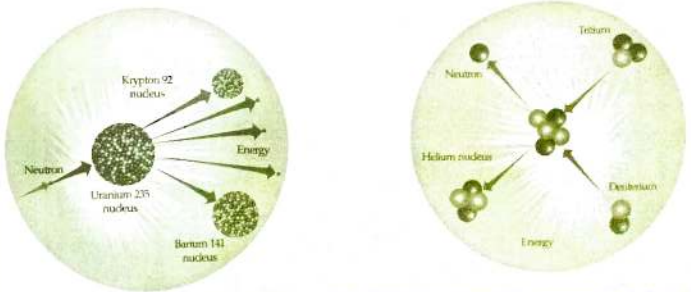
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## ISOTOPES




Every atom of an element contains the same number of protons. For example, Carbon has 6 protons. But sometimes the number of neutrons in elements varies and this results in isotopes. Sometimes, Carbon atoms have 8 neutrons instead of 6 and we get Carbon 14, an isotope of Carbon 12.

## THE FACTS BEHIND FISSION AND FUSION




Fission occurs when one element breaks into several others, releasing energy in the process. For example: When a neutron strikes Uranium it disintegrates into Krypton, Barium, a few neutrons and lots of energy. This energy is used by nuclear reactors. In fusion, two elements fuse together to form another element and also release energy. For example: Tritium and Deuterium fuse to form Helium and release enormous amounts of energy in the bargain! That's fission and fusion for you!

Source : Internet

# The Periodic Table

*Remembering the order of the first 10 elements of the periodic table is child's play!*
*Just memorize the sentence given below. The chemical symbol of each of these elements is given in bold type.*

**H**appy **He**nry **Li**kes **Be**ans, **B**rownies and **C**hocolate **N**uts **O**ver **F**riday's **Ne**ws

| 1 | 2 | | | | | | | | | | | 13 | 14 | 15 | 16 | 17 | 18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 H hydrogen | | | | | | | | | | | | | | | | | 2 He helium |
| 3 Li lithium | 4 Be beryllium | | | | | | | | | | | 5 B boron | 6 C carbon | 7 N nitrogen | 8 O oxygen | 9 F fluorine | 10 Ne neon |
| 11 Na sodium | 12 Mg magnesium | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 Al aluminium | 14 Si silicon | 15 P phosphorus | 16 S sulphur | 17 Cl chlorine | 18 Ar argon |
| 19 K potassium | 20 Ca calcium | 21 Sc scandium | 22 Ti titanium | 23 V vanadium | 24 Cr chromium | 25 Mn manganese | 26 Fe iron | 27 Co cobalt | 28 Ni nickel | 29 Cu copper | 30 Zn zinc | 31 Ga gallium | 32 Ge germanium | 33 As arsenic | 34 Se selenium | 35 Br bromine | 36 Kr krypton |
| 37 Rb rubidium | 38 Sr strontium | 39 Y yttrium | 40 Zr zirconium | 41 Nb niobium | 42 Mo molybdenum | 43 Tc technetium | 44 Ru ruthenium | 45 Rh rhodium | 46 Pd palladium | 47 Ag silver | 48 Cd cadmium | 49 In indium | 50 Sn tin | 51 Sb antimony | 52 Te tellurium | 53 I iodine | 54 Xe xenon |
| 55 Cs caesium | 56 Ba barium | 57-71 lanthanoids | 72 Hf hafnium | 73 Ta tantalum | 74 W tungsten | 75 Re rhenium | 76 Os osmium | 77 Ir iridium | 78 Pt platinum | 79 Au gold | 80 Hg mercury | 81 Tl thallium | 82 Pb lead | 83 Bi bismuth | 84 Po polonium | 85 At astatine | 86 Rn radon |
| 87 Fr francium | 88 Ra radium | 89-103 actinoids | 104 Rf rutherfordium | 105 Db dubnium | 106 Sg seaborgium | 107 Bh bohrium | 108 Hs hassium | 109 Mt meitnerium | 110 Ds darmstadium | 111 Rg roentgenium | 112 Cn copernicium | | | | | | |

Key: atomic number, Symbol, name

| 57 La lanthanum | 58 Ce cerium | 59 Pr praseodymium | 60 Nd neodymium | 61 Pm promethium | 62 Sm samarium | 63 Eu europium | 64 Gd gadolinium | 65 Tb terbium | 66 Dy dysprosium | 67 Ho holmium | 68 Er erbium | 69 Tm thulium | 70 Yb ytterbium | 71 Lu lutetium |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 89 Ac actinium | 90 Th thorium | 91 Pa protactinium | 92 U uranium | 93 Np neptunium | 94 Pu plutonium | 95 Am americium | 96 Cm curium | 97 Bk berkelium | 98 Cf californium | 99 Es einsteinium | 100 Fm fermium | 101 Md mendelevium | 102 No nobelium | 103 Lr lawrencium |

## COMMON NAMES OF DIFFERENT CHEMICALS

| Name | Common Chemical Name | Formula |
|---|---|---|
| Blue Vitriol | Copper sulphate | $CuSO_4.5H_2O$ |
| Green Vitriol | Ferrous sulphate | $FeSO_4.7H_2O$ |
| Epsom | Magnesium sulphate | $MgSO_4.7H_2O$ |
| Plaster of Paris | Calcium sulphate hemihydrate | $CaSO_4.\ 1/2\ H_2O$ |
| Gypsum | Calcium sulphate dihydrate | $CaSO_4.2H_2O$ |
| Washing soda | Sodium carbonate | $Na_2CO_3$ |
| Common salt or Rock salt | Sodium chloride | NaCl |
| Nitre or Salt Petre | Potassium nitrate | $KNO_3$ |
| Baking soda | Sodium bicarbonate | $NaHCO_3$ |
| Chalk | Calcium carbonate | $CaCO_3$ |
| Caustic lunar | Silver nitrate | $AgNO_3$ |
| Quick lime | Calcium oxide | CaO |
| Slaked lime or Milk of lime | Calcium hydroxide | $Ca(OH)_2$ |
| Potash Alum | Potassium aluminium sulphate dodecahydrate | $K_2SO_4Al_2(SO_4)_3.12H_2O$ |
| Bleaching powder | Calcium oxychloride | $CaOCl_2$ |
| Carbolic Acid | Phenol | $C_6H_5OH$ |
| Talc | Magnesium silicate | $MgSiO_3$ |
| Water glass | Sodium silicate | $Na_2SiO_3$ |

## CONVERSION FACTORS TO SI UNITS

| Physical quantity | CGS | SI |
|---|---|---|
| Force | 1 dyne | $= 10^{-5}$N(newton) |
| Pressure | 1 Ba | = 0.1 Pa (Pascal) |
| Energy | 1 erg | $= 10^{-7}$J(Joule) |
| Heat | 1 cal | = 4.148J |
| Volume | 1 Litre | $= 0.001\ m^3$ |

## STRENGTH OF AQUEOUS SOLUTION OF SOME CONCENTRATED ACIDS AND BASES

| Compound | Molecular formula | Molecular mass | Approximate | | | Volume required to make 1ℓ of approximate 1M solution (ml) |
|---|---|---|---|---|---|---|
| | | | % of solute by weight | Specific gravity | Molarity | |
| Hydrochloric acid | HCl | 36.5 | 36 | 1.18 | 11.65 | 85.8 |
| Nitric acid | $HNO_3$ | 63.0 | 70 | 1.42 | 15.8 | 63.3 |
| Sulphuric acid | $H_2SO_4$ | 98.0 | 98 | 1.84 | 18.4 | 54.3 |
| Acetic acid | $CH_3COOH$ | 60.0 | 99.5 | 1.05 | 17.5 | 57.5. |
| Phosphoric acid | $H_3PO_4$ | 98.0 | 85.5 | 1.71 | 14.8 | 67.5 |
| Sodium hydroxide | NaOH | 40.0 | 50.5 | 1.53 | 19.4 | 51.5 |
| Ammonia | $NH_3$ | 17.0 | 25.1 | 0.91 | 13.4 | 74.6 |

## WEIGHT OF SUBSTANCES REQUIRED TO PREPARE ONE LITRE OF A SOLUTION OF A PARTICULAR NORMALITY

| Sr. No. | Substance | Mol. Mass | Eq. Mass | Quantity in grams for solution of strength | | | |
|---|---|---|---|---|---|---|---|
| | | | | N/10 | N/20 | N/30 | N/40 |
| 1 | Oxalic acid, $H_2C_2O_4.2H_2O$ | 126.08 | 63.04 | 6.3040 | 3.1520 | 2.1013 | 1.5760 |
| 2 | Hydrochloric acid, HCl | 36.50 | 36.50 | 3.650 | 1.8250 | 1.2167 | 0.9125 |
| 3 | Sulphuric acid, $H_2SO_4$ | 98.00 | 49.00 | 4.9000 | 2.4500 | 1.6333 | 1.2250 |
| 4 | Acetic acid, $CH_3COOH$ | 60.00 | 60.00 | 6.0000 | 3.0000 | 2.0000 | 1.5000 |
| 5 | Sodium carbonate, $Na_2CO_3$ | 106.00 | 53.00 | 5.3000 | 2.6500 | 1.7667 | 1.3250 |
| 6 | Sodium hydroxide, NaOH | 40.00 | 40.00 | 4.0000 | 2.0000 | 1.3333 | 1.0000 |
| 7 | Sodium bicarbonate, $NaHCO_3$ | 84.00 | 84.00 | 8.4000 | 4.2000 | 2.8000 | 2.1000 |
| 8 | Calcium carbonate, $CaCO_3$ | 100.00 | 50.00 | 5.0000 | 2.5000 | 1.6667 | 1.2500 |
| 9 | Potassium dichromate, $K_2Cr_2O_7$ | 294.20 | 49.03 | 4.9030 | 2.4515 | 1.6343 | 1.2257 |
| 10 | Potassium permanganate, $KMnO_4$ | 158.03 | 31.60 | 3.1600 | 1.5800 | 1.0533 | 0.7900 |
| 11 | Mohr's salt $FeSO_4(NH_4)_2SO_4.6H_2O$ | 392.10 | 392.10 | 39.2100 | 19.6050 | 13.0700 | 9.8025 |
| 12 | Sodium thiosulphate, $Na_2S_2O_3.5H_2O$ | 248.20 | 248.20 | 24.8200 | 12.4100 | 8.2733 | 6.2050 |

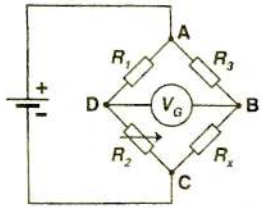
### Wheatstone bridge

In the figure, $\mathbf{R_x}$ is the unknown resistance to be measured; $\mathbf{R_1}$, $\mathbf{R_2}$ and $\mathbf{R_3}$ are resistors of known resistance and the resistance of $R_2$ is adjustable. If the ratio of the two resistances in the known leg ($R_2/R_1$) is equal to the ratio of the two in the unknown leg ($R_x/R_3$) then the voltage between the two midpoints (**B** and **D**) will be zero and no current will flow through the galvanometer $\mathbf{V_g}$. If the bridge is unbalanced, the direction of the current indicates whether $R_2$ is too high or too low. $R_2$ is varied until there is no current through the galvanometer, which then reads zero.

At the point of balance, the ratio of

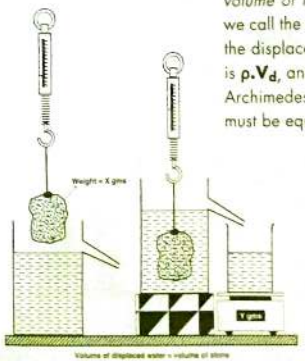
$$\frac{R_2}{R_1} = \frac{R_x}{R_3}$$

$$\therefore R_x = \frac{R_2}{R_1} R_3$$




### Archimedes Principle

To determine the magnitude of the buoyant force that a liquid exerts on a submerged object, we must consider how much of the object (of total volume V) lies below the liquid surface. Since only the submerged portion of the object displaces liquid, we see that *the displaced volume of liquid is equal to the object's submerged volume*. If we call the difference in density of the fluid and object as **ρ** and the displaced volume $\mathbf{V_d}$, then the mass of the displaced liquid is $\mathbf{\rho.V_d}$, and the *weight* of the displaced liquid is $\mathbf{\rho.g.V_d}$. From Archimedes' principle, the magnitude of the buoyant force B must be equal to this weight:

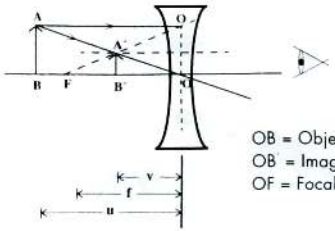



$$B = \rho.g.V_d$$

For an object *completely* submerged in the liquid, $V_d = V$. However, for an object only *partially* submerged, $V_d < V$.

## Focal length for concave lens

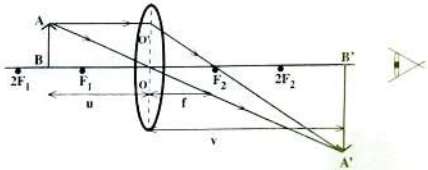
Consider an object **A B** placed at a distance of **u** in front of a concave lens of focal length " **f** " on the principal axis of the lens. Concave lens forms a virtual and erect image **A'B'** at a distance of " **v** " from the optical centre of the lens as shown in the diagram below.




OB = Object distance = u
OB = Image distance = v
OF = Focal length = f

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

## Focal length of convex lens




Let **AB** represent an object placed at right angles to the principal axis at a distance greater than the focal length **f** of the convex lens. The image **A'B'** is formed beyond $2F_2$ and is real and inverted.

OB = Object distance = u
OB = Image distance = v
$OF_2$ = Focal length = f

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

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# Introduction to Haematology

Haematology is defined as the branch of medical science which deals with the study of blood.

The word haematology is derived from Greek words:

- Haima - meaning blood
- Logos/Logia - meaning study.

It is primarily concerned with the study of formed elements of blood. Blood is the fluid constituent of the body that flows through vascular channels. The total volume of blood in a 70kg healthy adult is about 5.5L or about 7-8% of body weight. About 45% of this amount is composed of formed elements. Plasma constitutes about 5% of body weight.

## # Collection of blood samples

### Introduction

Blood is one of the most common specimen used in laboratory investigations. Venous blood is preferred for most haematological examinations.

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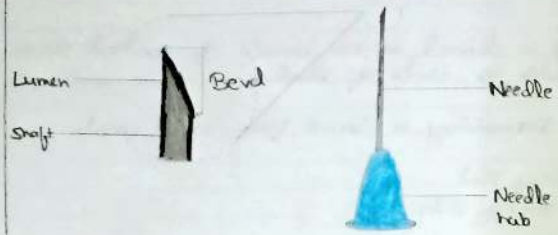



Fig. 24-gauge Needle

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The peripheral sample, i.e capillary blood, can be used satisfactorily for many purpose if free flow of blood is obtained. But this procedure should be avoided in patients who may be carrying transmissible diseases.

Capillary blood is commonly used for estimation of haemoglobin, cell count, blood grouping, bleeding and clotting time and other investigations that use less amount of blood, whereas venous blood is preffered for more comprehensive haematological investigation like ESR (Erythrocyte Sedimentation Rate), PCV etc.

## Method of Collection

For physiology practical, blood is often collected by pricking the tip of finger. Therefore collection of peripheral blood by puncture method is described below.

Principles

There are two general sources of blood for clinical laboratory tests:

1. Peripheral blood (capillary blood)
2. Venous blood.

• For small quantities of blood for haematological investigation the specimen is obtained from the capillary bed by

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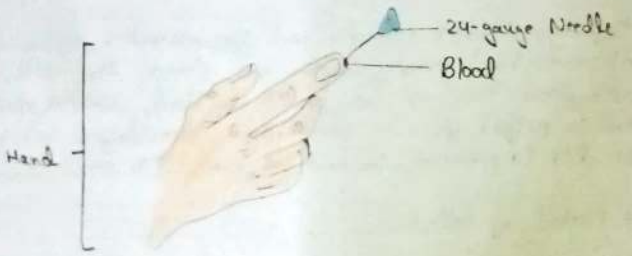



Fig. Pricking Finger

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puncturing the skin. The tip of the finger is the most common site for puncture.

- For larger quantities of blood, the puncture is made directly into the vein using a sterile syringe and needle collecting tube.

## 1. Collection from capillary blood

The capillary blood is often used for bedside investigations. But this blood is likely to give erroneous results if not collected properly.

Therefore, it should be used when it is not possible to obtain venous blood. Free flow of blood is essential and only gentle squeezing is permissible. Ideally, a long drop of blood is allowed to form spontaneously.

The blood can be obtained from the earlobe or fingertip of adults and older children, or from the heel/ plantar surface of the heel in infants.

### A) Pricking Finger

Use a sterile disposable lancet or needle of 24 gauge. In the traditional practical method, the needle may be sterilized by heating over a flame.

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Spirit (70% alcohol) kills most bacteria and viruses, including hepatitis B virus, provided the surface remains wet for adequate contact time. However, heating over flame is a more definitive method of sterilization.

However, the sharpness of the needle is adversely affected by heat and not by spirit.

Steps:

1. Digital pad of the second and third fingers on its palmer surface, about 3-5 mm lateral from the nail bed, is used for pricking purpose.

This area is less sensitive; therefore the prick is less painful.

2. Apply spirit to the fingers with cotton swab and do not prick till the spirit dries up completely.

Sterilization with spirit is effective after it has dried sufficiently; wet spirit should not come in contact with the blood sample.

3. Prick the finger by a single stabbing action, just deep enough (about 4-5 mm) to give free flow of blood.

4. Then wipe away the first few drops and collect the sample when blood flows continuously.

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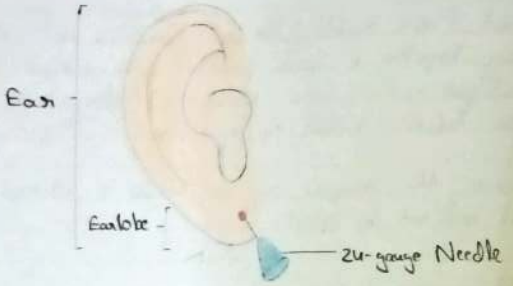



Fig. Pricking of the earlobe




Fig. Plantar surface of foot with puncture site.

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B) Pricking of the earlobe

Rub the earlobe in between the fingers until warm. Then prick it up to 2-3mm with a sterile lancet/needle by a single stabbing action. Wipe away the first few drops and collect the sample when blood flows continuously.

c) Pricking the heel

Pricking the heel is a satisfactory method to obtain a blood sample in infants. Make sure the heel is really warm. Then a deep prick is made into the lateral or medial part of the plantar surface of the heel. Wipe away the first few drops and collect the sample when the blood is flowing continuously.

2. Collection of Venous blood

The venous blood is preffered for most haematological examination. It is best obtained from an antecubital vein by means of dry disposable plastic bags/syringe. In obese patients, it may be easier to use a vein in the dorsum of the hand.

Procedure:

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1. Allow the subject to sit comfortably and support the arm on the edge of the table.

2. Apply the sphygmomanometer cuff, kept as a diastolic BP cuff or tourniquet to the upper arm so as to block venous return.

Ask the subject to open and close the fist repeatedly.

Clean the skin with spirit over the selected vein and allow it to dry, when the vein becomes suitable for puncture.

3. Place the thumb of the left hand about 5cm below the vein and apply traction on the skin towards the digits.

This fixes the vein and prevents it from slipping about during insertion of the needle.

4. By means of a dry disposable plastic syringe (23 or 24 gauge), the needle is inserted into the vein. When the needle enters the vein, all resistance trying to stop the blood will cease and blood will also be seen entering the barrel of syringe.

The piston of the syringe should withdrawn slowly and evenly and no attempt should be made to withdraw blood faster than the vein allows.

5. Loosen the tourniquet, withdraw the needle and apply pressure for several minutes till the bleeding stops.

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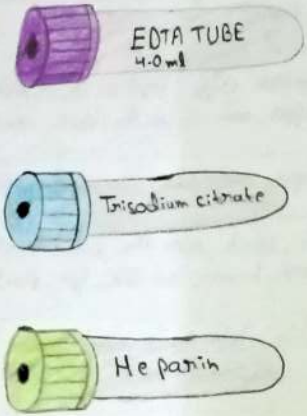



Fig. Commonly used anti-coagulants

Date

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6. After detaching the needle, deliver the blood carefully into the containers.

# Commonly used anti-coagulants

1. EDTA (Ethylene diamine tetra-acetate)
2. Trisodium citrate
3. Double oxalate
4. Heparin (naturally occuring anticoagulant)

1. EDTA: The sodium and potassium salts of EDTA are powerful anticoagulants. It acts by binding metal ions and by chelating calcium ions in the blood.

2. Trisodium citrate: It is an anticoagulant of choice in coagulation studies.

3. Double oxalate: This is an anticoagulant containing ammonium oxalate and potassium oxalate, hence it is called double oxalate.

Potassium oxalate causes shrinkage of RBCs, whereas ammonium oxalate increases their volume. Both combined use therefore helps to balance these effects.

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# Compound Microscope

## Parts of microscope

1. Stand
2. optical system
3. Body
4. Stage
5. Sub-stage
6. Mirror

## * Stand

→ Stand consist of heavy foot like horse shoe it provide stability and limb which bears the optical system. the limb is attached to the foot by a single hinge joint so that microscope may be set at comfortable angle for the observer.

## * Optical system

→ The optical system is mounted on tube which is usually in two parts.

1. External Tube - Which bears at its lower end, the revolving

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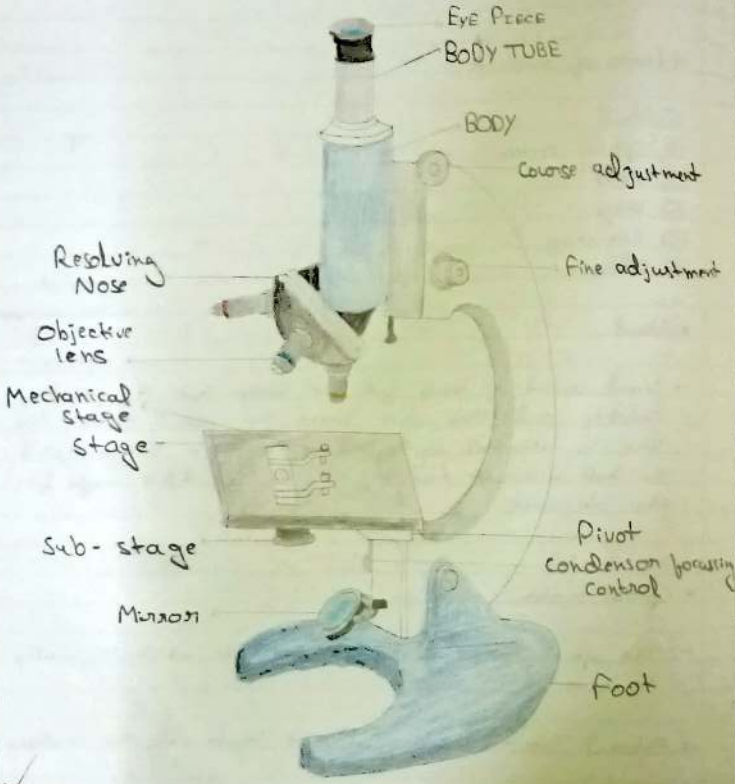



Fig. A COMPOUND MICROSCOPE

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nose piece is which interchangable three objective lenses of various magnification are fitted.

There lenses are: ① lowest power (×10)
② higher power (×40)
③ oil immersion (×100)

The revolving nose piece allows any objective lense to be brought into position below the body tube its correctd position being indicated by a soft click.

An inner draw tube that carries the eye piece (magnification-ral) at its upper end.

* Body

The whole assembly is held in a position by the body which consist of 2 mechanisms:

① Coarse
② Fine adjustments

with these adjustment the height of the tube can be adjusted so that the objective lens can be positioned at its optimal working distance (i.e. its focal length) from the object to be examined. The fine adjustment is usually

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graduated in 1/50th and each division corresponds to a movement of 0.002 mm of the tube, it is used for accurate focusing.

* Stage

The stage is a plat form that accomodates a glass slide on which the object to be examined is mounted. It is attached to the limb, immediately below the level of the objective lens and has an aperture in the centre to permit light to reach the object.

A mechanical stage is fitted to the fixed stage so, that the object may be moved from side to side and forward-backward

* Sub-stage

→ It lies beneath the stage which can be powered or raised by means of a screw. The stage is fitted with a condenser and iris diaphragm.

(i) The condenser consists of two lenses. It serves to condense the rays of light and to focus them on the object. The height of condenser and therefore, the focus of light can be varied.

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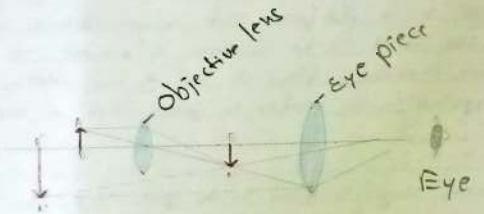



Fig: Image formation in compound Microscope

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(ii) Immediately below the condenser is the iris diaphragm that can control the amount of light reaching the object

*Mirror

→ The fitted to the tail piece below the condenser is a double reflecting mirror, which is plane on one side and concave on the other. The plane side of mirror is used for when the light source is natural or day light (distance)

*Formation of image

→ It is the lenses comprising the objective which initiate the magnifying process. An objective decreases at a distance from the object that roughly equal it to its focal length, and admits rays that are to form a real, an inverted, enlarged image in the upper part of the tube. At this point there is interposed another lens, the field lens whose function is to collect the diverging rays of the primary image so that they may pass through the eye lens of the eye piece which magnifies the image still further.

The field lens is accommodated in the lower plane of the eye piece. Rays as they leave the eye piece to reach the observers eye, are once more divergent and thus, the image

Teacher's Signature

seen by the retina is visited which appears to be about 25 cm (10 inches) in front of the eyes.

*Magnification

⇒ The magnification of the object is determined by multiplying the powers of the eye piece with that of the objective (x inscribed on the objective or eye piece denotes the unit of the magnification)

For example: if the eye piece magnification is ×10 and that of objective lens is ×100, than a given object magnified by ×1000.

* Resolution

⇒ The limit of useful magnification is set by the resolving power of the lens i.e its ability to reveal clearly the adjacent structural details as separate and distinct. It is this power which determines the amount of structural details that can be observed.

• Oil immersion objective



Fig. Objective lenses

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⇒ Oil immersion objective is most frequently used in bacteriology because of its greater magnification and resolution.

It has been seen that any factor which reduces the amount of rays falling on the objective, impairs the quality of the image in case of oil immersion objective, the space between the object and the objective here is filled with some transparent fluid (such as liquid paraffin, glycerine or transparent oil) with the same refraction index as glass (1.515) then the rays of light do not undergo refraction and pass directly into the objective. The most common used fluid is the cedar wood oil.

# Focussing of an object

* Under Low Power

1) Place the slide to be focused on the stage and fit it into the mechanical stage. By means of the screws of the mechanical stage (Bring the object on the central aperture in the stage).

2) Using the light source and by adjusting the plane mirror, focus the light on the object, partially close the iris diaphragm and lower the objective (x10) with coarse adjustment near the slide.

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3) The position of the slide is so adjusted that the aperture of the objective lens is just above the point to be focussed on the slide.

4) Now look into the eye-piece the optical tube is moved upwards (again using coarse adjustment) slowly and cautiously till the object comes under view. Once this is achieved, the coarse adjustment should not be used.

5) Lowers the condenser from the top position till the object is properly illuminated. The condenser usually occupies a low position.

6) Next, using the fine adjustment only, the slide can be brought to a sharp focus.

B. Under High Powers

1) To focus under high power (×40) simply rotate the nose-piece so that the high power objective lens may come to live 1cm above the slide in a sharp focus.

2) The iris diaphragm is fully open and concave mirror is used to condense the light on the object.

3) The condenser is kept in the beginning at the top and

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is then gradually lowered till the object is properly illuminated. The condenser usually occupies a higher position than that to focus under the low power.

4.) With the fine adjustment screw, attempt the final focusing.

C) Under oil immersion lens

1) To focus under oil immersion lens (×100), a drop of cedar-wood oil is placed on the slide.

Then the nose piece is further turned so that oil immersion lens may be in use, it will automatically clip into the drop of oil.

2) As the slide is moved, the drop of oil will also get moved on its own. Thus, any part of the slide can be observed.

3) The iris diaphragm is kept fully opened and the condenser is kept in the top position.

# Precautions

1) A clear image can never be seen through dirty lens. Make sure that the eye piece as well as all the objectives are clean.

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While cleaning microscope lenses do not rub them with dry cotton or rough cloth which might leave permanent scratch marks on the lenses. Apply Xylene on the lenses (to dissolve the grease) and then wipe dry gently with a piece of silk cloth or softest material available.

2. Do not clean the objective lenses with alcohol, as it may dissolve the cement which unites the component lenses.

3. Never loosen the optical tube while you look through the eye piece.

4. Make Minimal and careful use of only the fine adjustment while using the high power and oil immersion lenses.

5. If the microscope has to be moved, it should be held upright by the limb keeping a hand below its foot.

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## Estimation of Haemoglobin

**Aim:-** To estimate Hb of your own blood sample

**Apparatus:-**

(i) Haemoglobinometer
(ii) Comparator box
(iii) Graduated haemoglobinometer tube
(iv) Haemoglobin pipette
(v) Glass rod
(vi) Pricking needle
(vii) Stop watch

**Chemical**

(i) N/10 HCl
(ii) Distilled water
(iii) Spirit, cotton swap

**Method:-**

Sahli's haemoglobinometer consist of:

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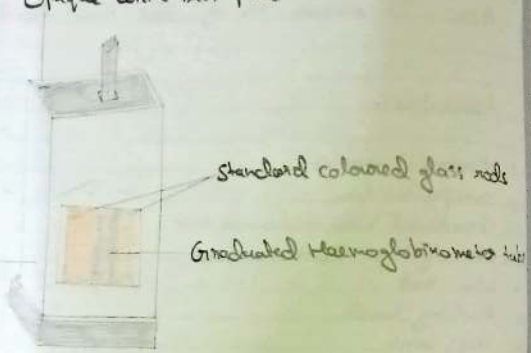



Fig. Haemoglobinometer

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1) A single mark pipette (haemoglobinometer pipette) measures 20 cu (cmm) of blood

2) A round or square haemoglobinometer tube graduated in percentage (10 to 140) gram per cent (2 to 22) square tube is preferred because error due to curved tube is avoided.

3) A glass rod which is used as a stirrer

4) A rectangular box, comparator fitted with two solid standard coloured glass rods. The opaque white background provides the necessary contrast for easy comparison.

Principle :-

The amount of haemoglobin can be estimated by the conversion of known of volume of blood into acid haematin by the addition of dilute HCl and subsequent colorimetric comparison with a suitable standard.

Procedure :-

1) With the help of a dropper, take N/10 HCl in the graduated haemoglobinometer tube upto its lowest mark (usually, 10% or 2gm%)

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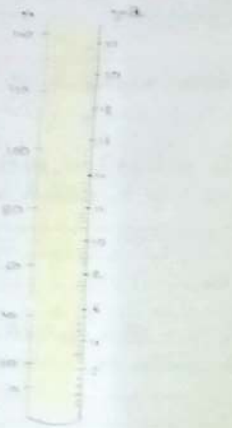


Fig- Graduated haemoglobinometer tube

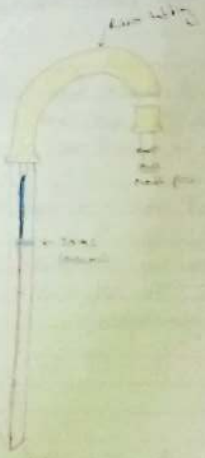


Fig- Haemoglobinometer pipette

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2) Prick the fingers under aseptic precautions when a drop of reasonable size has collected, hold the pipette horizontally, apply its tip to the drop and draw exactly 20 µl of blood into the pipette, taking care that there should be no air bubbles. Take the pipette away and wipe off any blood which adhere to the tip.

3) Immediately transfer the blood from the pipette into N/10 HCl in the graduated haemoglobinometer tube and rinse the pipette several times by drawing the N/10 HCl acid to mix the blood. Avoid foaming.

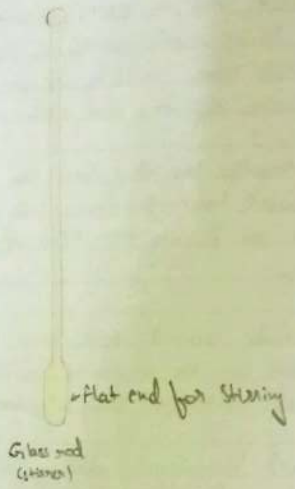
4) Mix the contents thoroughly and leave the solution to stand for about 10 minutes for the maximum conversion of haemoglobin in blood to acid haematin (brown in colour)

5) Dilute the acid haematin by adding distilled water in drops. keep mixing it with the stirrer thoroughly.

6) Continue dilution till the colour matches with that of the standard. Before comparing the colour, the stirrer should be lifted above the solution only.

Note the reading of the meniscus from the scale provided on the haemoglobinometer tube and express the haemoglobin content as gm per 100ml of blood

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Precautions

1) The fingers should not be squezed to take out the blood otherwise tissue fluid might come out thus giving the lower estimation of haemoglobin.

2) Blood from the pipette should be transferred quickly into the tube containing N/10 HCl to prevent the clotting of blood in the pipette.

3) Blood, after mixing with N/10 HCl, should be left at least 8-10 minutes for complete formation of acid haematin.

4) Distilled water should be added drop by drop to dilute the acid haematin, otherwise over dilution occurs.

5) The matching of colours should be done only against the natural source of light.

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| Blood Type | Anti-A | Anti-B | Anti-D | Control |
|---|---|---|---|---|
| O-positive | | | | |
| O-Negative | | | | |
| A-positive | | | | |
| A-Negative | | | | |
| B-positive | | | | |
| B-Negative | | | | |
| AB-positive | | | | |
| AB-Negative | | | | |
| Invalid | | | | |

Fig ABO Blood Grouping System

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## Determination of Blood Group (ABO)

**Aim:** To determine blood group of your own sample by using ABO system

**Apparatus:** (i) Anti-serum kit contains Anti-A, Anti-B, Anti-D
(ii) Glass slide
(iii) 0.9% saline solution
(iv) Microscope
(v) Pricking needle
(vi) Glass marking pencil
(vii) Glass rod

**Principle -**

The red cell contain different types of Agglutinogen and plasma contain agglutinin. The red cell of subject are allowed to react with commercially made agglutinin. The presence and absence of clumping in different agglutinin determines the blood group

**Method -**

1) Cell suspension: For preparation of cell suspension (0.9% NaCl) add one drop of blood mix it properly

2) By using direct blood sample.

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Procedure:

1. Make a sterile finger prick and collect a large drop of blood on glass slide.

Before taking the Blood sample you have to make glass slide of "Anti-A", "Anti-B", "Anti-D" and control.

2. Place a drop of Anti-A serum on its left half where label a Anti-A with the help of glass marking pencil on other hand of the another place drop of Anti-B serum where label as Anti-B

3. Place a drop of saline on right hand of each slide and mark as control.

4. To each of its drop add a drop of red cell suspension or direct blood sample by using capillary pipette.

5. Mix the red cell suspension with Anti-serum using seperate glass rod or seperate wooden stick by gently shaking and wait for 5-10 minutes.

6. Observe the serum cell mixture for agglutination reaction (Clumping) compare it with cell saline control.

7. Confirm your findings under the low power objective (10x)

8. The serum cell mixture should be examine before it dries up from nonformation of red cell.

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| Antigen-A | Antigen-B | Rh-Antigen | Blood group |
|---|---|---|---|
| Absent | Absent | Present | $O^{+ve}$ |

9) Record the note down the presence and absence of agglutination on each slide and interpret the result.

Result: Blood group of my own blood group is ..$O^{+ve}$..

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## BLEEDING TIME

**AIM:** To determine bleeding time of your own blood sample.

**Apparatus:** Pricking needle, whatmann filter paper (1 cm²), stopwatch

**Method:** Duke Method -

⇒ This method is the most frequently used method to determine BT in rat laboratories. As it is easy to perform and require minimal equipment and laboratory scale.

**Principle:**

⇒ Bleeding time is the time taken from the puncture of blood vessels to the stopping of bleeding. A deep skin puncture is made and the length of the time required for the bleeding is recorded. It determine the function of platelets and integrity of the capillaries.

**Procedure:**

1. Assemble all necessary material and give proper instructions to the subject
2. Set the stopwatch at zero mark
3. Clean the finger tip of the subject with spirit/alcohol and

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allow the skin to dry completely.

4. Make a deep puncture with the help of sterile lancet or prick needle.

5. Immediately start the stopwatch and note the time of puncture it is referred to as zero time (or 0 seconds)

6. Blot the drop of blood coming out of the incision every 30 sec by using the blotting paper or whatman filter. Place each subsequent drop a little farther along the side of the filter paper.

7. Repeat the procedure every 30 sec using fresh area of paper on each occasion until bleeding stops and further blood spot appear on the each blot of the blood on the filter paper represent 30 of blood.

8. Count the total no. of blood spot on the filter paper and multiply by ½. This gives bleeding time in minutes. Normal bleeding time by this method is 2-6 min.

Result:

The bleeding time of my own blood sample is ~~normal~~ 2.5 min.

Conclusion: My bleeding time is normal

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Precaution:

(1) Gather all the necessary material before starting to test

2. Do not rub the finger too much while cleaning the skin as rubbing increases the blood flow

3. The skin should be completely dry

4. Do not squeeze the finger

Clinical significance

⇒ Prolongation of bleeding time is found in thrombocytopenia and ~~bone marrow disease~~ von willebrand factor (vWF)

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= 0.9 mm diameter

fig. Capillary filled with blood

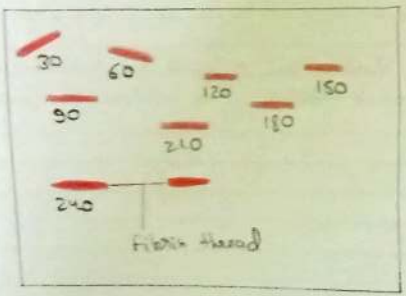



fig. Breaking the capillary every 30 secs. till a fibrin thread appears

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## Clotting time

**Aim:** To determine clotting time of your own blood sample.

**Apparatus:** Pricking needle, stopwatch, glass capillary tube.

**Method:** capillary glass tube method -

⇒ It is the most convinient and commonly used in laboratory.

**Principle:**

⇒ The clotting time is the time taken from the pricking of finger upto the formation of fibrin thread. A standard incision is made in the skin of the patient and blood is taken into the capillary glass pipette. The length of the time it takes for the blood to clot is reported as clotting time.

**Procedure:**

⇒ Explain the procedure to the subject, make a sterile finger puncture by using the lancet upto a depth of 3-4 mm.

2. As soon as the blood is visible, start the stopwatch or at the time of finger pricking start the stopwatch.

3. When a large drop of blood has collected, introduce one end

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of capillary tube into the drop holding the tube such that the other end is at a relatively lower end level. The blood enters rapidly into the capillary tube by capillary action mechanism.

4. Hold the capillary tube filled with blood in the palm of hand so as to keep it at maintained body temperature.

5. At the end of 1 min. Break off about 1cm of the tube from one end and notice if a thread of fibrin connect to the broken end of the tube.

6. If there is no fibrin thread, repeat the procedure every 20s till the fibrin thread appears.

7. The total time taken from the puncture till the formation of thread is called the clotting time.

8. The normal time of clotting blood is 2-8 min.

Clinical significance:

The clotting time is prolonged in diseases in which clotting factors are deficient. If the CT is more than 10 min, the patient should be subjected to detail investigation to identify the missing coagulation factor.

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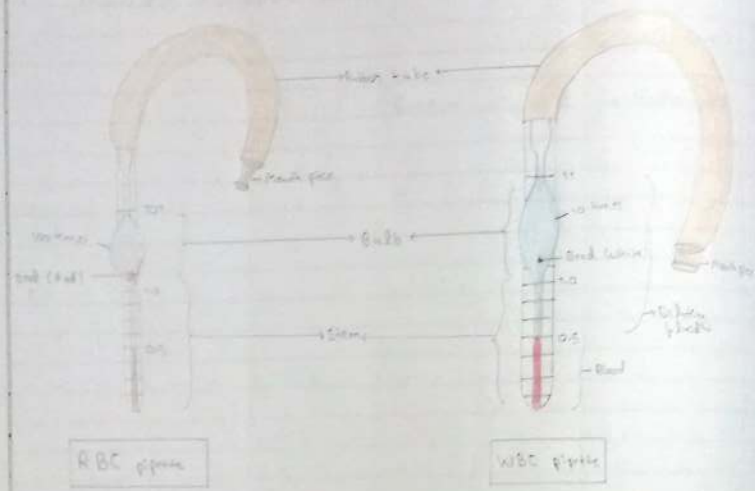
Result:

The clotting time of my own blood sample is 4 ~~3-5~~ minutes

Conclusion:

My clotting time is normal

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# Haemocytometry

Aim: Study of haemocytometer

Haemocytometer is an apparatus to perform count of various blood cells that is RBC, WBC, Blood platelets. It consists of RBC pipette, WBC pipette, Neubauer's counting chamber.

A → Pipette:

Pipette is used to dilute the blood with a suitable fluid, in an accurately known proportion. Since, RBCs are more numerous than WBCs different dilutions have to be made for the RBC and WBC counts.

Hence, 2 pipettes are provided.

1. RBC pipette:

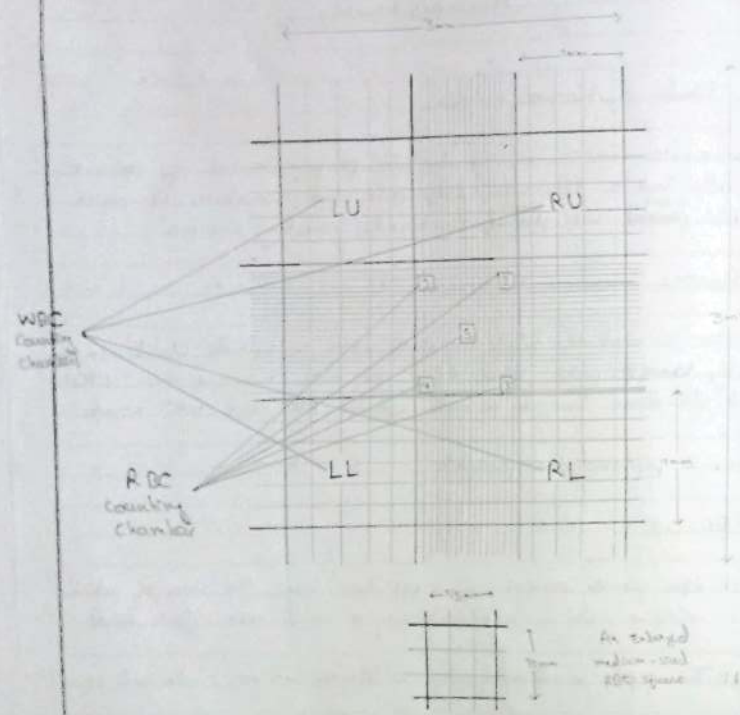
(I) RBC pipette consist of a capillary tube, the stem of which widens into a bulb containing a small mix glass bead.

(II) The pipette is calibrated into 10 division between bulb and tip.

(III) The centre calibration being marked 0.5 and nearest the bulb 1.0.

(IV) On the opposite side of the bulb is another calibration 101

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Improved Neubauer's chamber

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The marking refered to relative rather than absolute volume

(IV) The pipette is supplied with a rubber tube and mouth piece (rubber teat) to facilitate filling by suction. These rubber tubing is sufficiently long to enable the observer to read the pipette calibrations comfortably during the process of filling

2. WBC pipette:

(I) WBC pipette is similar to RBC pipette but has a smaller bulb with a white glass bead. The graduations are 0.5 and 1 on the stem and 11 above the bulb

B → Improved Neubauer counting chamber:

The counting chamber was originally developed in 1885 and was later modified by Neubauer.

(1) It is used to obtain very thin film of the fluid of the known volume for cellular count.

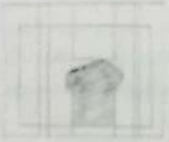
(II) It consist of a thick glass slide with a polished central platform divided by a short transverse gutter into two portions. Each portion is ruled with a counting grid

(III) On either side, the central platform is bounded by a groove

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called mast each mast extend, is bounded on its outside by another platform, which is slightly higher than their central platform.

(iii) Perfectly ground cover slip rests upon the four lateral platforms, these, bridging the moats are covering the central platform. The cavities is so constructed that there is a space of 0.1 mm between the ruled platform and the undersurface of cover slip.

(iv) The counting grid is made up of ruled area measure is 3 mm x 3mm. This area is divided by triple or double lines into 9 nine large equal squares each having an area of 1 sq.mm. Of these the four last squares at the corners are used for WBC counting while the leafe square in the center is used for RBC counting.

(v) Each of the four large squares at the corner of the ruled area is used for WBC counting and is called WBC square. Each square is further divided into 16 small squares, each having an area of 1/16 sq.mm.

(vi) The central large square is for RBC counting and is called RBC square. It has an area of 1 sq.mm. It is further divided by the help of triple or double line into 25 medium sized squares. Each with an area of 1/25 sq.mm. Each medium sized square is further divided into 16 small squares each with an area of 1/400 sq.mm.

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(VII) For the total RBC count the central square containing 400 smallest square is used. Each smallest square has an area of 1/400 sq.mm. If a cover slip is placed, the depth of the chamber is 1/10 mm making the total volume over each of the smallest square as 1/4000 mL.

(VIII) For total leukocyte count (TLC), 4 corner squares are used each of these corner squares has a length of 1mm and depth of 1/10 mm, so that the volume of ~~fluid~~ over 1 square is 1/10 mL

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## Determination of Total leucocyte count

**Aim:** Determination of total lymphocyte count

**Apparatus:** Improved Neubauer's counting chamber, WBC pipette, WBC diluting fluid, Turk's fluid, compound microscope, cover slip, pricking apparatus

**Composition of WBC diluting fluid:** Each 100ml of fluid contains -

1. Glacial acetic acid (2%): 3 ml (To destroy the WBC, RBC, Platelate membrane).
2. Gentian violet (1%): 1ml (To stain the nuclei of WBC deep violet black so that they may be identified easily.
3. Distill water: To make 100ml (solvent)

**Principle:**

Since, the WBC count is much lower than RBC count. So much lower dilution is required i.e. dilute 1 in 20 that is 1:20.

**Procedure:**

1. Take WBC diluting fluid in a watch glass. ~~after pr~~

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2. After pricking the finger suck the second drop of blood with the help of the pipette exactly up to exactly 0.5 mark and dilute it with WBC diluting fluid by sucking the fluid up to 11 mark (Dilution 1 is 20).

3. Gently rotate the pipette for atleast 3-4 minute. Note for 3-5 minute of the diluted blood.

4. A clean cover slip is placed on the Neubeuer's chamber. So, as to cover the ruled area of both sides.

5. Focus the WBC squares of Neubeuer's ruling under low power objective (x10) of the microscope. All WBC squares can be seen on microscope view. The chamber is then removed from the microscope then placed on the table.

6. Discard the few drops of the WBC fluid in the stem of the pipette. Charge the counting chamber and allow the time for settling of cells.

7. To charge the counting chamber a small drop of blood mixed with diluting fluid and is allowed to form a drop at the tip of the WBC pipette.

8. Bring the tip of the pipette at the edge of coverslip on the central platform and put the drop gently by making an angle of 45°.

Once the fluid is entered it uniformly spreads over the chamber.

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by capillary action.

8. Wait for 3-5 minutes for the settlement of cells. Place the counting chamber again under the microscope which is already focused at low power objective by using the fine adjustment screw. Look for uniform distribution of WBC on the large square located at 4 corner of counting chamber. These 4 corner square are used to perform total leucocyte count and known as WBC square.

WBC under low power objective appear as round refractile cells with dark violet dots. (nucleus).

Change the chamber again, if WBC distribution is found to be ununiformly distributed.

9. If distribution of WBC is found uniform in all 4 corner of WBC square that is left upper, right upper, left lower and right lower respectively. Count WBC in each of 4 corners of WBC square that are divided into 16 medium sized square. Each of having an area of 1/16 sq. mm.

10 While counting WBC dust particles in fluids can be counted as WBC but in contrast with WBC they are non-refractile.

Rule of counting:

We follow the 'inverted 'L' rule for avoiding count of same cell twice, Counting should be started from left and upper border.

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of square.

(ii) Cells lie with the square are counted in same square.

(iii) Any cell which is half of more than half are counted in same square

(iv) The cell lying on upper or left borders, are counted in the square.

(v) Omit the cell lying on lower or right border of square as they will be counted on next adjacent square.

Calculation of dilution factor in WBC pipette:

(i) $\frac{\text{Final volume achieved (10 parts)}}{\text{original volume taken (0.5 parts)}} = 20$

(2) Calculation of volume of fluid:

Area of 4 WBC square = 4 x 1mm x 1mm = 4 sq.mm
Depth of camber = 0.1mm

Therefore the volume of fluid over 4 WBC square is equal to 4 sq.mm x 0.1mm = 0.4 μL.

3. Calculation of total lymphocyte count:

Let the total no of WBCs in 4 WBCs square that is 0.4 μL of

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## Calculation

N = 207

Total WBC = 207 × 50/μL
= 10350/μL

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diluted blood. Then, total no of WBC in 1 μL of blood is

$$= \frac{N \times \text{Dilution factor (50)}}{0.4} \Rightarrow N \times 50$$

Result:

WBC count of my blood is 10350. /μL of blood.

Normal count

1) Normal total leukocyte count 4000–11000 /μL of blood
2) Average 7000 /μL of blood
3) In New born 15000–25000 /μL
4) In infants 6000–18000 /μL

Conditions leading to total leukocyte count greater than 11000 is called leucocytosis

Conditions that Alter total leucocyte count

1 Newborns and infants
2 Physical exercise.
3 After food intake
4 Expose to sun and increased environmental temperature
5 Parturition
6. New born

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Pathological

1) Acute inflammation caused by phytogenic
2) Chronic infection like TB
3) Trauma
4) Burns
5) Surgery
6) Myocardial

Condition leading to decrease in total WBC count lower than 4000 is called leucopenia

Pathological

1) Typhoid fever
2) Malaria
3) Kala-azar
4) Radiation (Chemotherapy)
5) Influenza
6) Viral or protozoal infection.
7) Bone marrow depression.

OPSE

1) Checks for pipette cleanliness, dryness
2) Take sufficient diluting fluid into a watch glass
3) Suck the blood into the pipette exactly up to the 0.5 mark and there are no air bubbles

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4.) Wipes off the blood which sticks to the tip of pipette.

5.) Sucks diluting fluid with the help of pipette gently upto the 11 mark, just above the bulb.

6.) Immediately hold the pipette in a horizontal position between the palms and

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4.) Wipes off the blood which sticks to the tip of pipette.

5.) Sucks diluting fluid with the help of pipette exactly upto the 11 mark, just above the bulb.

6.) Immediately hold the pipette in a horizontal position between the palms and

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# Determination of RBC count

**Aim:** Determination of total Erythrocyte count

**Apparatus:** Compound Microscope, Improved Neubauer's counting chamber, RBC pipette, RBC diluting fluid (Hayem's fluid), cover slip, Watch glass, pricking apparatus

## RBC Diluting fluid (Hayems fluid)

- **Composition:** Each 100ml of fluid contains ① Sodium sulphate - 2.5 gram to prevent aggregation of RBCs rouleax formation

② Sodium chloride - 0.5 gram, helps to maintain the isotonicity of the fluid. Therefore RBCs remain suspended in the solution

③ Mercury chloride - 0.25 gram, act as preservative and shows anti-bacterial and anti-fungal action

④ Distill water - 100ml act as solvent

## Priciple

Since the normal RBC count runs in million, the count is made possible by diluting blood sample before counting and subsequently multiplying the count by the Dilution factor.

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## Procedure

1. Ensure that improved Neubauer's chamber, cover slips and the base of the Microscope by absolutely clean and the RBC pipette is dry and patent.
2. Take adequate RBC diluting fluid in a watch glass.
3. Prick the finger taking the necessary precautions and obtain an optimal blood drop size.
4. Wipe off the first drop of the blood and hold the pipette horizontally. apply it's tip to the drop and suck next drop of blood in RBC pipette exactly upto 0.5 mark, taking care that there should be no air bubble.
5. Wipe off the blood sticking to the tip of pipette.
6. Immediately, thereafter, holding the pipette in a nearly horizontal position. RBC diluting fluid is sucked into the pipette exactly upto 101 mark just above the bulb.
7. The pipette is then placed horizontally into the palm and roll gently for about a minute to ensure the thorough mixing of the blood and the fluid. This gives a suspension of RBC in the bulb of the pipette.
8. A clean grease free cover slip is placed in the Neubauer's chamber

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So as to cover the ruled area

(9) Focus RBC counting square of counting chamber under low power objective lense of microscope. The chamber is then removed from the microscope and placed on the table

(10) Discard the first few drops of fluid from the RBC pipette

(11) Charging the Neubauer's chamber and follow the same procedure as done during total leukocyte count

(12) Allow some time from about 1 minute for cells to settle down in the counting chamber

(13) The charged Neubauer's chamber is now kept under low power of the microscope already in focused. Using only the fine adjustment the cell along the ruled area is brought into focus

(14) Check whether cells are uniformly distributed or not. If not clean the Neubauer's chamber and recharge it.

(15) Switch over to high power and focus the RBC squares under high power objective and count the RBCs in 4 medium sized corner squares and 1 medium sized central square

(16) Rules for the counting of cells follows the same procedure as done in total leukocyte count.

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Calculations

1. Calculation of Dilution factor

Dilution factor = $\frac{\text{Final volume achieved}}{\text{Original volume taken}}$

$\Rightarrow 100/0.5$

$\Rightarrow 200$

2. Calculation of volume of fluid examined

Area of 5 radian used RBC square = $5 \times \frac{1}{5}$ mm $\times \frac{1}{5}$ mm

$= \frac{1}{5}$ mm$^2$

Depth of the chamber = 0.1 mm

Hence, volume of fluid over 5 RBC squares $\Rightarrow \frac{1}{5}$ mm$^2 \times$ 0.1 mm

$\Rightarrow \frac{1}{50}$ µL

3. Calculation of total RBC count

Let N be the number of total RBC counted in 5 squares that is $\frac{1}{50}$ µL of blood. Therefore, No. of RBCs in 1 µL of undiluted blood $\Rightarrow N \times 50 \times$ Dilution factor (200)

$\Rightarrow N \times 10{,}000$

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## Calculation

N = 512

Total RBC = 512 × 10,000/cu.mm
= 5,120,000/cu.mm
= 5.12 million/cu.mm

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## Result

RBC count of my blood is 5.120... million/cu.mm of blood.

## Discussion

1) Normal RBC count in male : 5-6 million/cu.mm of blood
Normal RBC count in female : 4.5-5 million/cu.mm of blood
Normal RBC count in newborn : 6-7 million/cu.mm of blood.

2) Conditions results in increased RBC count.

① Polycythemia - RBC count is more than 6 million/cu.mm of blood.

### physiological cause

① Hypoxia
② High altitude
③ New born
④ Male > female
⑤ Stress and exercise

### Pathological cause

① Cancer in bone marrow (Poly
② Heart and lung disease
③ Dehydration
④ Thyrotoxicosis

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3) Conditions decreasing RBC count (Anemia):- RBC < 4 million/cu. HBC < 12gm/dL and PCV is less than 40%

Physiological cause of Anemia

(i) Three months to 1 year of life
(ii) Pregnancy.
(iii) Female < Male (RBC count)

Pathological cause - (Minimal).

OPSE

(1) Check that the pipette is absolutely dry and patent.

(2) Take adequat RBC diluting fluid.

(3) Suck the blood into pipette exactly upto 0.5 mark with No air bubble.

(4) Wipe off the blood sticking to the tip of pipette.

(5) Suck diluting fluid into the pipette exactly upto 101 mark.

(6) Immediately hold the pipette in horizontal position between palms and roll gently

OPSE 2.

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1) Ensure that the Improve Neubauer's chamber and cover slips are clean.

2) Properly mixed the diluted Blood sample by rolling Horizontally in the palm.

3.) Placed the cover slip on the Neubauer's chamber so as to cover the ruled area on both the sides.

4) Discard the first few drops of the fluid from the pipette.

5) Charged the chamber with a single adequate size drop which just fills the chamber without leaving any air bubbles or fluid flowing into the gutters.

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